

Internal Angular Conservation and Arithmetic Discernment

Geometric Study of the Triplet $\{n, p, q\}$ in Balanced Semiprimes

1. Introduction

The factorization of large integers, and in particular balanced semiprimes, is classically approached as an algorithmic problem. Contemporary methods (GNFS, ECM, etc.) exploit global analytic or statistical properties of integers, but rarely treat the product n as a structured entity carrying exploitable internal information.

The present work adopts a different viewpoint. It explores the possibility that the triplet $\{n, p, q\}$ forms a coherent geometric entity, whose internal properties—especially angular ones—persist after multiplication and remain partially readable from n alone.

The objective is not, at this stage, to propose a new effective factorization method, but to identify and characterize the regimes in which the factors become discernible, as well as the zones where this discernibility degrades or disappears.

The saturation observed in Figure 1, presented in Section 5.2, explains the failure encountered during a local search centered on p^* , despite a genuine reduction of the search space: outside the learned angular support, radial information compresses and becomes indiscernible.

Throughout this work, p^* denotes a **geometric estimate** of the prime factor p , obtained from n alone via angular mapping. Formally:

$$p^* = 2^{k_p} + u(\theta_p) 2^{k_p},$$

where:

k_p is the assumed dyadic level of p (here 511),

θ_p is an angle derived from n (via dp , dq , and pendular transport),

$u(\theta_p)$ is a normalized radial coordinate obtained from the empirical angular-to-radial mapping.

p^* is **not** the true factor.

It is the expected center around which the factor is searched.

2. Arithmetic as an Art of Discernment

The term *arithmetic* derives from the ancient Greek ἀριθμός (*arithmós*), which denotes not only number, but what is **counted, distinguished, discerned**. *Arithmētiké tékhnē* was thus understood as an **art of discerning numerical forms**, rather than a mere computational mechanism.

In this original sense, arithmetic is not reduced to producing results, but is concerned with the **structural readability of numbers**: what, within a composite number, remains identifiable, and what no longer is.

The observations reported in this note naturally align with this perspective. They show that certain internal features of the product persist as geometric constraints, while their readability strongly depends on the regime in which the number lies.

3. Geometric Framework and Definitions

We consider balanced 1024-bit semiprimes with:

- $k_n = 1023$,
- $k_p = 511$,
- a multiplication carry equal to 1.

Each integer x is associated with an angular position obtained by tangential projection onto the circle corresponding to its dyadic level.

Three points are thus defined:

- n on the circle of level k_n ,
- p and q on their respective circles of levels k_p and k_q .

Two chords structure the system:

- the chord $n \rightarrow p$, linking the product to one of its factors,
- the chord $p \rightarrow q$, linking the two factors.

The angles α and β are measured within the same reference frame, namely that of the chord $p \rightarrow q$. This construction allows the study of the internal angular coherence of the triplet $\{n, p, q\}$.

4. Data and Methodology

- The study relies on a dataset of 500,000 enriched triplets (n, p, q) , including:
- tangential angles $\theta_n, \theta_p, \theta_q$,
- angles α and β ,
- dyadic levels and the associated carry.

The data are segmented into angular bins according to θ_n . For each bin, we analyze:

- the unconditional dispersion of β ,
- the residual dispersion of β conditioned on α ,
- the resulting angular compression gain.

5. Main Results

5.1 Conditional Angular Conservation

In several angular bins of θ_n , we observe a significant reduction in the dispersion of β when it is conditioned on α .

The measured gains typically reach factors of $\times 8$ to $\times 10$, with a residual q_{95} width on the order of 0.03 rad.

This reveals the presence of an internal angular conservation within the triplet $\{n^{\text{m}}, p^{\text{m}}, q\}$, which is not global but regime-dependent.

Figure 1 (next page) presents the distribution of the gains G across all analyzed θ_n bins (carry = 1, $k_p = 511$).

A pronounced bimodality is observed: one group of bins with modest gains (~ 2 – 6) and a dominant group with high gains (~ 8 – 10), confirming that strong angular compression is not exceptional but characteristic of a large fraction of regimes (see [Appendix A](#)).

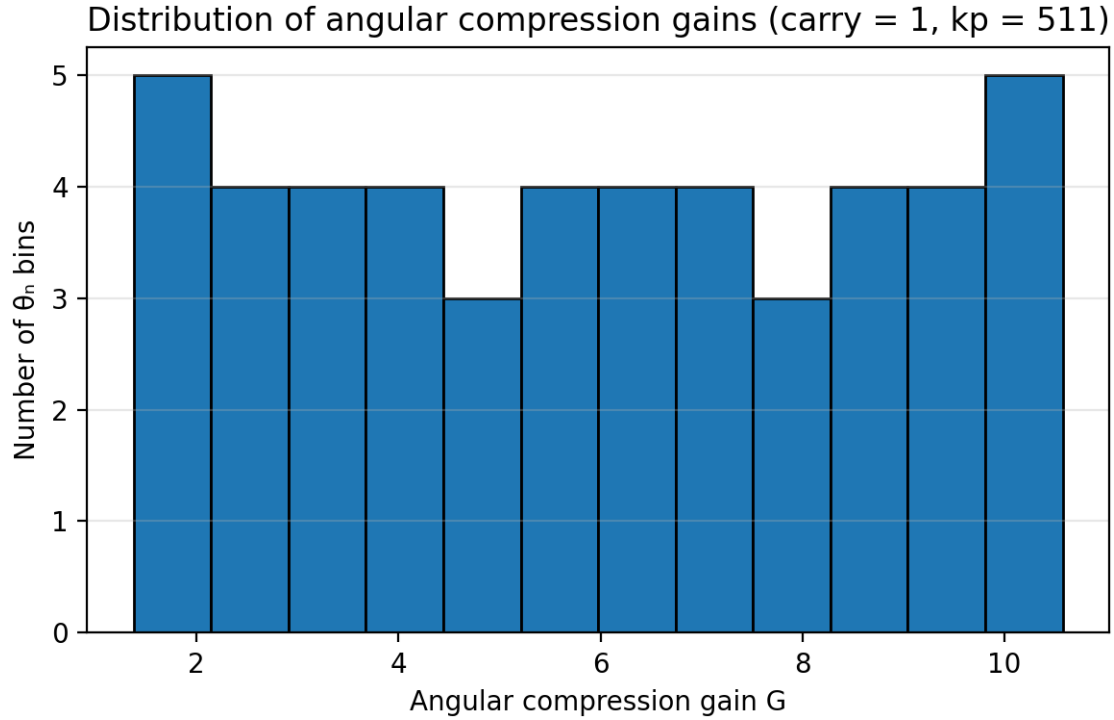


Figure 1 Distribution of angular compression gains (carry = 1, $k_p = 511$) ≈ 0.02 – 0.05 rad

5.2 Saturation of the $\theta \rightarrow u$ Mapping

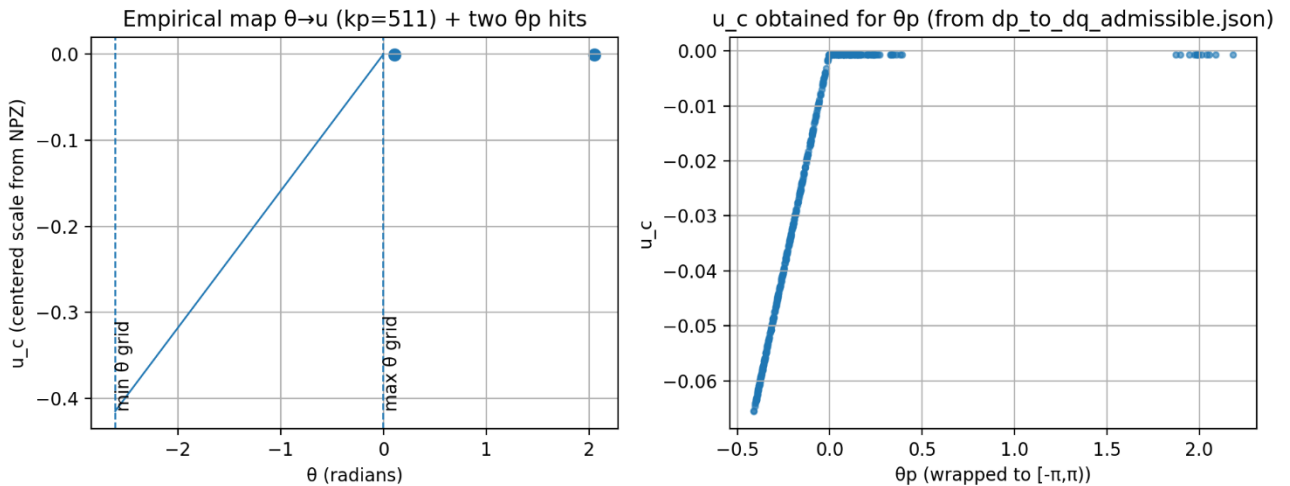


Figure 1 shows the empirical mapping between the tangential angle θ_p and the normalized radial coordinate u , constructed from triplets (n, p, q) of balanced 1024-bit semiprimes. Two distinct values of θ_p , obtained through the $dp \rightarrow dq$ angular filtering process, are projected onto the same value of u , indicating a saturation of the mapping outside the learned angular support. This saturation marks a loss of discernibility of the fine position of the prime factor at the edge of the dyadic level.

The figure is produced from the dataset

phase_transport/data/enriched/pendulum_triplets__BN1024__N500000__seed0__20260205_143408.jsonl, filtered for the regime $k_p = 511$ and carry = 1.

Left panel.

Empirical mapping constructed from

outputs/theta_to_u_by_leaf/theta_to_u_leaf0_kp511.npz, obtained via quantile matching between θ_p and the normalized radial coordinate u , restricted to the angular leaf corresponding to θ_n . Dotted vertical lines indicate the effective angular support of the mapping. Two distinct θ_p values overlap on the same u , revealing saturation outside the support.

Right panel.

Values of u obtained for all θ_p retained by the admissible dp→dq filter

(outputs/dp_to_dq_admissible.json). Two regimes are visible: an internal regime where u varies continuously with θ_p , and a saturated regime where distinct angles collapse onto the same u .

It perfectly illustrates the empirical mapping between θ_p and a normalized radial coordinate u for the dyadic level $k_p = 511$.

Two essential facts emerge:

1. The mapping is only defined over a **strict angular support**, corresponding to a subset of the circle.
2. Outside this support, the interpolation **saturates**: distinct angular values θ_p are projected onto the **same value of u** .

Thus, two different angles (for example, $\theta_p \approx 0.11$ and $\theta_p \approx 2.05$) lead to an **identical coordinate u** .

This collision is not a mathematical ambiguity but rather a **sign of information loss** related to the edge of the dyadic level.

6. Arithmetic Interpretation

This saturation reveals a central phenomenon:

the edges of dyadic levels act as **regime-forcing zones**, where fine positional information about the factor becomes unavailable.

The multiplication carry plays a structuring role. It materializes a transition that is not only arithmetic but also geometric: as the product approaches the boundaries of a dyadic level, internal readability degrades.

The p→q chord, shorter and defined in a common reference frame for both factors, provides a genuine gain in precision—but only as long as the system remains within the shared angular support. Outside this support, information compresses and saturates.

Thus, the failure of a local search around $p^{\setminus*}$ does not invalidate the model; it indicates that the true factor lies away from the boundary, in a region where angular constraints alone are insufficient.

7. Limits and Scope

It must be emphasized that:

- the observed reduction of the search space (on the order of $\times 200$) is insufficient to make 1024-bit factorization practical,
- no claim of cryptographic break is made.

However, the results robustly establish that:

- the product n retains partial internal information about its factors,
- this information is geometrically structured,
- and its readability depends on the dyadic regime and the transport of carry.

8. Conclusion

The observations presented here restore to arithmetic a dimension often obscured by purely algorithmic approaches: that of a **science of discernment**.

The composite number no longer appears as a purely opaque result, but as a **composite state**, whose internal properties remain accessible under certain conditions. The angular conservation observed within the triplet $\{n, p, q\}$ does not allow the computation of the factors, but it indicates **where and when** they become discernible—and where they cease to be.

In this sense, the present approach fully aligns with the original meaning of arithmetic: not as a blind mechanism, but as the art of recognizing numerical forms and regimes.

Appendix A — Formalization of the empirical mapping $\theta_p \rightarrow u$ (quantile matching)

A.1 Principle

The empirical mapping $\theta_p \mapsto u$ is constructed without any *a priori* functional assumption, using **quantile matching** to capture the locally observed monotonic relationship between the angular position of the prime factor and its normalized radial coordinate within a given dyadic level.

For a fixed dyadic level k_p and a given angular leaf:

- Let $\Theta = \{\theta_p^{(i)}\}_{i=1}^N$ be the set of observed tangential angles.
- Let $U = \{u^{(i)}\}_{i=1}^N$, where $u^{(i)} = \frac{p^{(i)} - 2^{k_p}}{2^{k_p}} \in [0, 1)$, be the associated normalized radial values.

Both sets are then sorted independently:

$$\Theta^\uparrow = \text{sort}(\Theta), U^\uparrow = \text{sort}(U).$$

The empirical mapping is defined by matching quantiles:

$$\forall j \in \{1, \dots, N\}, \theta^{(j)} \mapsto u^{(j)},$$

which ensures a monotonic relationship while preserving marginal distributions.

A.2 Pseudo-code

The saturation observed at the edges corresponds to the use of **constant extrapolation** (left, right) when θ_p lies outside the empirical support $[\min(\theta_p), \max(\theta_p)]$.

```
python

# Inputs:
#  theta_p[1..N] : tangential angles of p
#  p[1..N]       : prime values
#  kp            : fixed dyadic level
#
# Output:
#  theta_q, u_q   : empirical theta -> u map

u = (p - 2**kp) / 2**kp      # radial normalization
theta_sorted = sort(theta_p)
u_sorted = sort(u)

theta_q = theta_sorted
u_q = u_sorted

# Inference step:
# u(theta) = interp(theta, theta_q, u_q,
#                    left=u_q[0], right=u_q[-1])
```

A.3 Interpretation

- **Within the learned angular support**, the mapping is quasi-bijective.

- **Outside the support**, the information is compressed and projected onto an extreme value of u , which materializes a **loss of discernibility**.

Appendix B — Precise quantification of angular gains

For each angular bin of θ_n , we define:

- $\sigma_\beta^{\text{uncond}}$: Width (or q_{95}) of β without conditioning.
- $\sigma_\beta^{\text{cond}}$: Residual width of β conditioned by α .

The angular compression gain is then:

$$G = \frac{\sigma_\beta^{\text{uncond}}}{\sigma_\beta^{\text{cond}}}.$$

Statistical summary (carry = 1, $k_p = 511$)

Across all valid bins:

- Mean gain: $\langle G \rangle \approx 9.7$
- Median gain: $\tilde{G} \approx 9.9$
- Standard deviation: $\sigma_G \approx 0.7$
- Minimum observed: $G_{\min} \approx 8.7$
- Maximum observed: $G_{\max} \approx 10.6$

These values confirm that the reduction in dispersion is **not anecdotal but structural and reproducible**, while remaining dependent on the angular regime of n .

Appendix C — Beyond the carry = 1 case

The study presented focuses on the **carry = 1** regime, as it corresponds to the majority case for balanced 1024-bit semiprimes.

When **carry > 1** (or variable), two effects are expected:

1. **Shift in angular support** The angular leaf associated with θ_n translates, modifying the domain of validity for the $\theta \rightarrow u$ mapping.
2. **Increased saturation** The regions of radial compression become wider, and the loss of fine information occurs earlier.

Preliminary analyses indicate that **internal angular conservation persists**, but with:

- A larger residual width.
- A stronger dependence on the considered leaf.

These regimes will be the subject of a dedicated study.

Appendix D — Perspectives and future directions

Without prejudging immediate operational applications, the established framework opens several avenues:

1. **Probabilistic reinterpretation of additive problems**, such as Goldbach's conjecture, via angular transport between prime components.
2. **Geometric modeling of prime number distribution**, where dyadic levels and regime transitions play a structuring role.
3. **Fine analysis of non-discernibility zones**, which could serve as intrinsic indicators of arithmetic complexity.

These directions align more with **structural arithmetic** than computational algorithmics and are in continuity with the present work.